Omkar Shettigar

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Professional Summary

Data Scientist & AI/ML Engineer with 2 years of experience in sustainabile finance domain, specializing in production-grade LLM systems, Agentic AI, and RAG retrieval pipelines. Experienced in architecting hybrid decision engines that blend deterministic logic, dense neural embeddings (Sentence-BERT, FAISS), and generative LLM reasoning (Ollama, LLaMA, Gemma). Proven track record of taking technical leadership to stabilize complex ML pipelines, eliminating model drift through rigorous ground-truth evaluation harnesses, and optimizing high-throughput feature pipelines processing 300,000+ records at scale.

Education

Visvesvaraya Technological University

B.E., Artificial Intelligence and Data Science, GPA: 8.44 / 10.0 *Jan 2021 - Jan 2025*

Skills

Programming Languages: Python, JavaScript, C, C#, TypeScript, Java

Machine Learning & NLP: Sentence-BERT (SBERT), FAISS (Vector Indexing & Search), PyTorch, Scikit-learn, XGBoost, Char-TF-IDF, Supervised Classification, Confidence Calibration, Human-in-the-Loop (HITL) ML, Model Evaluation & Parity Testing, Feature Engineering, PII Masking

Databases: Vector Databases, PostgreSQL, MySQL, Supabase, Turso

ML Systems & Performance: High-Throughput ML Pipelines, Model Serving (FastAPI, Docker), Model Monitoring (Drift/Latency/Quality), Inference Profiling (cProfile), Aho-Corasick Automata, LRU Caching, MLflow, A/B Testing

Cloud Platforms: AWS, EC2, S3, RDS, Sagemaker, Vercel, Supabase, GCP, Azure

Generative AI & LLMs: LangChain, LangGraph, Agentic AI Architectures, Retrieval-Augmented Generation (RAG), Ollama, Hugging Face, Prompt Engineering (Few-Shot, Structured JSON Outputs), Hallucination Mitigation, MCP Protocol, LLaMA-3, Gemma

Work Experience

Data Scientist Mar 2025 - Present

Catalysk *Bengaluru*

- Built a hybrid semantic retrieval (RAG) system integrating Sentence-BERT (all-MiniLM-L6-v2) and FAISS L2 vector indexing across 17,500+ merchant entities and MCC embeddings; augmented LLM context with multi-signal payment metadata and reduced neural search latency from 436s to <25s through vector batching and embedding memorization.
- Stepped up to lead technical stabilization when peer architectural merges introduced severe regressions and inference bottlenecks; engineered a 4,535-sample Bit-for-Bit Golden Parity & Decision Trace Harness validating model predictions across 7 categorical and feature dimensions, establishing a zero-drift CI/CD gate.
- Architected an end-to-end Agentic AI & progressive classification funnel for financial intelligence, orchestrating multi-tier routing (Deterministic Rules → Dense Vector Retrieval → Few-Shot LLM Reasoning via Ollama); engineered structured JSON schemas and hallucination-mitigation maps to ensure 100% adherence to financial taxonomy.
- Developed observable, production-grade AI services (AWS EC2 + S3) with automated A/B testing on spending decile variables; achieved 85% classification accuracy improvement on sustainability scoring engine (Logistic Regression + Peer Group Normalization); reduced manual reporting effort by 60%.
- Optimized the end-to-end ML feature extraction and inference pipeline across 300,000+ consumer records; eliminated $O(N^2)$ orchestrator bottlenecks and redundant feature transformations, cutting runtime by 50% (2160s to 125s at ~2449 records/sec) with a 6.8x speedup in classification stage latency (156s to 23s).
- Engineered a high-precision ML counterparty classification engine (P2P vs. P2M) standardizing 600+ banking narration patterns (UPI/IMPS/NEFT); designed multi-layer orthographic and semantic guardrails to separate personal Indian names/initials from commercial merchants, eliminating a 39% false-positive misclassification rate.
- Designed a composable skill-pack system for merchant classification: vector-DB-backed skill modules with MCC keyword embeddings and LLM fallback routing with confidence scoring, boosted classification accuracy from 60% to high 70s, surfacing low confidence results for analyst review

Machine Learning Engineer Intern Nov 2023 - Dec 2023

SM Khannukar Associates

- Built LSTM/RNN time-series forecasting pipeline for stock prediction achieving 85% directional accuracy over 30-day backtest; automated ingestion via Yahoo Finance API, reducing preprocessing time by 40%.
- Executed statistical hypothesis testing for model validation; applied feature engineering and ensemble methods to improve signal quality and reduce overfitting.

Projects

Ancient Modern Apparel - Fashion Platform

- Built full-stack fashion-tech product (Next.js 15, React 19, TypeScript, Tailwind CSS) with Python FastAPI backend for AI-driven 4-view (front/back/left/right) avatar generation and 3D garment preview flows.
- Architected REST API routes for garments, avatars, and designs; integrated Three.js / React Three Fiber for real-time 3D rendering; Drizzle ORM + Turso for scalable DB ops and Supabase for hosted asset storage. Shipped auth, session management, personalized catalog + cart, and CI/CD deployment on Vercel.

Knee Osteoarthritis Detection using CNN — CISC-2024

- Designed 'Compact Vision Net' achieving 90% accuracy across 13 X-ray cancer classes, outperforming ResNet and MobileNet; 45% accuracy improvement via transfer learning and domain-specific fine-tuning.
- Integrated LLaMA-3 powered RAG chatbot (LangChain + Groq) for evidence-based clinical recommendations; applied CLAHE, Gamma Correction, Gaussian smoothing preprocessing for image quality enhancement.
- Integrated LLaMA-3 powered RAG system for evidence-based medical recommendations, reducing LLM dependency through retrieval-augmented generation over peer-reviewed literature.
- Presented research findings at international conference (CISC-2024), demonstrating model effectiveness and clinical applicability to peer reviewers and domain experts.